

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Pranay_

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.

A.

Cloud Architecture Design

The cloud architecture should include:

- Cloud Load Balancer
- Auto Scaling Groups
- Content Delivery Network (CDN)
- Object Storage for video files
- Virtual Machines or Containers for streaming services
- Cloud Database for user information
- Monitoring and Logging services

Load Balancing and Auto-Scaling

Load Balancing

- Distributes user requests across multiple servers.

- Prevents server overload.
- Improves response time.

Auto-Scaling

- Automatically increases servers during peak traffic.
- Removes unnecessary servers during low traffic.
- Ensures uninterrupted streaming.

Caching Techniques

- Use CDN to cache videos near users.
- Store frequently accessed videos in cache.
- Reduce latency and buffering.

Cost Optimization Strategies

- Use Auto Scaling to avoid paying for idle resources.
- Store old videos in low-cost archive storage.
- Use Reserved Instances for predictable workloads.
- Compress video files to reduce storage and bandwidth costs.

Conclusion

This architecture provides high scalability, reduced buffering, improved user experience, and cost-efficient cloud resource utilization.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

A.

Proposed Solution

- Package applications using Docker containers.
- Manage containers using Kubernetes.
- Deploy clusters across multiple cloud providers.
- Use Infrastructure as Code (IaC) for consistent deployments.

Container Orchestration

Kubernetes provides:

- Automated deployment
- Load balancing
- Self-healing
- Rolling updates
- Automatic scaling

Service Discovery

- Kubernetes automatically assigns DNS names.
- Services communicate using internal DNS.
- No manual IP configuration is required.

Scaling

- Horizontal Pod Autoscaler increases or decreases containers.
- Cluster Autoscaler adds or removes worker nodes.

Benefits

- High availability
- Vendor independence
- Disaster recovery

- Better scalability
- Improved reliability

Risks

- Increased management complexity
- Higher networking costs
- Security policy differences
- Data synchronization challenges

Conclusion

A Kubernetes-based multi-cloud architecture provides reliable, scalable, and portable application deployment.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.

The organization needs a robust backup and disaster recovery strategy.

Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

A. Backup and Disaster Recovery Design

- Enable automatic cloud backups.
- Store backup copies in different regions.
- Enable object versioning.
- Use cross-region replication.
- Schedule periodic backups.

Recovery Time Objective (RTO)

RTO is the maximum acceptable downtime.

To achieve a low RTO:

- Maintain standby servers.
- Automate recovery.
- Use replicated storage.
- Perform regular recovery testing.

Recovery Point Objective (RPO)

RPO is the maximum acceptable data loss.

To achieve a low RPO:

- Perform frequent backups.
- Enable continuous replication.
- Use database snapshots.

Business Continuity Steps

1. Detect failure.
2. Restore data from backup.
3. Redirect users to standby servers.
4. Verify system functionality.
5. Resume normal operations.

Conclusion

Cloud-native backup, replication, and versioning ensure fast recovery, minimal downtime, and protection against accidental data loss.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

A.

Hybrid Cloud Architecture

- Sensitive applications remain on-premises.
- Non-sensitive applications are hosted in the public cloud.
- Connect both environments using a secure VPN or dedicated private connection.
- Manage resources through centralized monitoring.

Identity Management

- Implement Single Sign-On (SSO).
- Use Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Integrate with a centralized identity provider.

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Secure encryption keys using a Key Management Service (KMS).

Challenges

- Complex infrastructure management
- Compliance requirements
- Latency between environments
- Security policy consistency
- Higher operational costs

Conclusion

A hybrid cloud offers flexibility while maintaining security and regulatory compliance for sensitive government data.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

A.

Multi-Region Deployment Strategy

- Deploy the application in multiple cloud regions.
- Use global load balancing to direct traffic.
- Maintain active-active or active-passive deployment.

Data Replication

- Replicate databases across regions.
- Use synchronous replication for critical data.
- Use asynchronous replication for distant regions.

Failover Mechanism

- Monitor application health continuously.
- Automatically redirect traffic if one region fails.
- Maintain standby resources in secondary regions.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network usage.
- Track application performance and availability.
- Configure alerts for failures and unusual activity.
- Maintain centralized logs for troubleshooting.

Benefits

- High availability
- Fault tolerance
- Reduced downtime
- Improved disaster recovery
- Better customer experience

Conclusion

A multi-region cloud deployment with replication, automatic failover, and continuous monitoring ensures uninterrupted service even if an entire region becomes unavailable.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented